

STAIR: Per-Problem Discrete Structure in Test-Time Compute Scaling for LLM Reasoning

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Abstract

Test-time compute scaling—allocating more reasoning tokens to improve LLM accuracy—is widely modeled as a smooth, monotonic, task-determined function. We present evidence challenging all three assumptions. We introduce STAIR (Staircase Test-time Adaptive Inference Routing), a framework that decomposes population-level scaling curves into per-problem components via BIC model selection and optimizes compute allocation through complexity-stratified temperature tuning. Through controlled experiments on a synthetic reasoning channel (800 problems, 4 factorial conditions, 5 seeds) and real LLM inference (Qwen2.5-0.5B and Qwen2.5-1.5B on GSM8K, 5 token budgets $\times$ 3 temperatures $\times$ 4 samples/cell), we find: **(1)** 87.5% of real per-problem scaling curves are better fit by piecewise-constant (staircase) functions than smooth sigmoids under BIC; **(2)** accuracy non-monotonicity—where more reasoning tokens *reduce* accuracy—occurs in 17–18% of real problem-model pairs; and **(3)** sequential computation depth predicts scaling elbows far more accurately than description-length proxies ($\rho = 0.96$ vs. $\rho = 0.38$ on synthetic data; gzip-step-count $\rho = 0.22$ on GSM8K). On synthetic data, the STAIR allocator reduces elbow prediction MAPE by 35% relative to five baselines, with gains concentrated on high-complexity problems. We prove that under mild conditions on the per-problem critical depth distribution, the population scaling curve is log-concave even when individual curves are discrete, formalizing the “smooth population from discrete individuals” phenomenon.

1 Introduction

The deployment cost of large language models is increasingly dominated by inference. Reasoning-intensive systems—OpenAI’s o-series [13], DeepSeek-R1 [4], Claude with extended thinking—generate thousands of chain-of-thought (CoT) tokens before a final answer. Understanding *when to stop thinking* is economically urgent: a 30% reduction in average token budget at constant accuracy directly translates to 30% lower serving cost.

Prior work models the test-time compute–accuracy relationship as a smooth, monotonically increasing curve with diminishing returns [9, 17]. Information-theoretic treatments frame CoT as iterative channel coding, deriving bounds on the mutual information between reasoning traces and correct solutions [14]. These bounds predict a log-concave scaling curve whose “elbow point” t^* depends on the task’s Kolmogorov complexity $K(x)$.

Three critical assumptions remain untested:

1. **Smoothness.** Does the log-concave curve hold *per problem*, or only in population expectation?
2. **Task-determination.** Is the elbow set by the task, or does the model’s “channel quality” dominate?
3. **Kolmogorov complexity.** Is description length the right complexity measure, or does *computational* depth matter more?

We introduce STAIR, a framework that addresses all three through a combination of theoretical analysis and empirical validation. Our contributions:

1. **Per-problem staircase decomposition (§6.1).** On real LLMs (Qwen2.5-1.5B, GSM8K), 87.5% of per-problem accuracy curves are better described by piecewise-constant functions than smooth sigmoids under

BIC model selection—far exceeding the 41% rate observed in our synthetic channel. We prove (Theorem 1) that a population of 1-staircase individuals produces a smooth population curve under log-concave critical depth distributions.

2. **Model-dependent channel capacity** (§6.2). accuracy non-monotonicity (“reasoning overshoot”) occurs in 17–18% of real problem-model pairs. Cross-model elbow divergence is 8.8% between Qwen-0.5B and Qwen-1.5B.
3. **Computational vs. description complexity** (§6.3). Circuit depth predicts scaling elbows with $\rho = 0.96$ (synthetic), while gzip compression achieves only $\rho = 0.38$. On real GSM8K, gzip–step-count $\rho = 0.22$, confirming the distinction.
4. **STAIR allocator and baselines** (§6). We compare against 5 baselines on synthetic data. STAIR achieves 47.1% MAPE vs. 72.1% for the best baseline, a 35% relative improvement concentrated on high-complexity problems (27.6% vs. 92.6%).

2 Related Work

Scaling laws. Training-time scaling laws [7, 9] are well-established power-law relationships between compute, data, parameters, and loss. Test-time analogs are newer: Snell et al. [17] showed test-time compute scaling can outperform parameter scaling; Brown et al. [1] demonstrated scaling in extended-thinking models. Jones [8] analyzed scaling laws through board game experiments. All model the curve as smooth.

Information theory and reasoning. Polyanskiy and Wu [14] provide channel coding foundations. Feng et al. [5] formalized CoT as iterative refinement with information-theoretic guarantees. Li and Vítányi [12] established the Kolmogorov complexity framework. The information bottleneck [3] connects compression to prediction. STAIR extends this line by introducing model-specific channel quality and non-monotonicity detection.

Adaptive inference. Early exit [6, 16], speculative decoding [11], mixture-of-depths [15], and best-of- N sampling all allocate inference compute adaptively. These approaches use learned routers with inference-time overhead. STAIR’s allocator uses a pre-inference complexity proxy (gzip compression for bucketing) with $O(n)$ overhead and zero model forward passes for routing.

3 Theoretical Framework

Definition 1 (Per-problem accuracy function). *For problem x with correct solution y^* , the accuracy function $a(t \mid x) = \Pr[y^* \mid r_t, x]$ maps token budget t to the probability of a correct answer given reasoning trace r_t of length t .*

Definition 2 (k -Staircase function). *$a(t \mid x)$ is k -staircase if there exist thresholds $\tau_1 < \dots < \tau_k$ and levels $\alpha_0 < \alpha_1 < \dots < \alpha_k$ such that $a(t \mid x) = \alpha_j$ for $\tau_j \leq t < \tau_{j+1}$. The critical depth $\tau(x) := \tau_1$ is the smallest budget where accuracy first increases.*

Definition 3 (Population elbow). *The population accuracy $\bar{a}(t) = \mathbb{E}_x[a(t \mid x)]$ and the population elbow $t^* = \arg \max_t |\bar{a}''(t)|$ is the point of maximum curvature change.*

Theorem 1 (Smooth population from discrete individuals). *Let each problem x_i have a 1-staircase accuracy function with critical depth τ_i , where $a(t \mid x_i) = \alpha_0$ for $t < \tau_i$ and $a(t \mid x_i) = \alpha_1$ for $t \geq \tau_i$. If the critical depths $\{\tau_i\}$ are drawn from a distribution with log-concave PDF f_τ , then the population accuracy $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0)F_\tau(t)$ is log-concave, where F_τ is the CDF.*

Proof. The population accuracy is $\bar{a}(t) = \alpha_0 + (\alpha_1 - \alpha_0)F_\tau(t)$, which is an affine transformation of $F_\tau(t)$. Since f_τ is log-concave, F_τ is log-concave by Prékopa’s theorem. An affine transformation of a log-concave function with positive coefficients is log-concave. Therefore $\bar{a}(t)$ is log-concave, and its second derivative $\bar{a}''(t) = (\alpha_1 - \alpha_0)f'_\tau(t)$ has a unique zero (the mode of f_τ), which defines the population elbow $t^* = \text{mode}(f_\tau)$. $\square$

Corollary 1. *The population elbow t^* equals the mode of the critical depth distribution, not any individual problem’s critical depth. The elbow is a property of the problem population, not any individual problem.*

Algorithm 1 STAIR Adaptive Inference Budget Allocation

Require: Problem x , gzip boundaries $[g_1, g_2]$, per-bucket temps $\{\tau_k^*\}$

- 1: $g \leftarrow |\text{gzip}(x, \text{level} = 9)| \{O(n) \text{ complexity proxy}\}$
 - 2: $k \leftarrow \text{tercile}(g; g_1, g_2) \{\text{Difficulty bucket: easy/med/hard}\}$
 - 3: $\tau^* \leftarrow \tau_k^* \{\text{Pre-calibrated per-bucket temperature}\}$
 - 4: Run BIC classification at temperature τ^*
 - 5: $\hat{d}_c \leftarrow$ critical depth from staircase fit (or midpoint from sigmoid)
 - 6: **return** Token budget $\hat{d}_c$
-

Proposition 1 (Accuracy non-monotonicity prevalence). *If a model has probability p of producing an accuracy-decreasing transition at each budget step (due to reasoning drift, truncation artifacts, or autoregressive error accumulation), then the probability of observing at least one non-monotonic step in a sequence of T budget levels is $1 - (1 - p)^T$. This simple model explains why non-monotonicity is more prevalent in long budget sweeps and with noisier models. **Note:** This concerns empirical accuracy, not mutual information—the data processing inequality is not violated.*

4 Method: STAIR Framework

4.1 Per-Problem BIC Classification

For each problem x_i and budget $t \in \{t_1, \dots, t_B\}$, we estimate accuracy $\hat{a}_i(t)$ from S independent samples. We fit two competing models:

Piecewise-constant: $\hat{a}_i(t) = \alpha_0 \mathbb{I}[t < \tau_i] + \alpha_1 \mathbb{I}[t \geq \tau_i]$ (2 effective parameters after grid search for τ_i).

Logistic sigmoid: $\hat{a}_i(t) = L / (1 + e^{-k(t-t_0)})$ with 3 parameters $\{L, k, t_0\}$.

We perform model selection via BIC. Following standard practice for small-sample model comparison [10], we compute $\text{BIC} = n \ln(\text{RSS}/n) + p \ln(n)$ and classify problem x_i as “staircase” if $\Delta \text{BIC} > 2$ (strong evidence).

Statistical note. We use MSE-based BIC for simplicity and transparency. A binomial likelihood model with logit link (GLM) would be more statistically appropriate for accuracy data derived from S Bernoulli trials per budget level. In Appendix A, we verify that likelihood-based BIC yields qualitatively similar staircase win rates (within 5pp), and we report both MSE-based and likelihood-based results. The key finding—that the majority of real per-problem curves favor discrete over smooth fits—is robust to this choice.

4.2 Cross-Model Divergence

For models m_A, m_B , we compute per-problem elbows and measure divergence:

$$\Delta_{\text{model}} = \frac{1}{N} \sum_{i=1}^N \frac{|t_{m_A}^*(x_i) - t_{m_B}^*(x_i)|}{\max(t_{m_A}^*(x_i), t_{m_B}^*(x_i))} \quad (1)$$

We detect *accuracy non-monotonicity*: cases where $\hat{a}_i(t_{j+1}) < \hat{a}_i(t_j) - \epsilon$ with $\epsilon = 0.15$ (chosen to exceed 2σ of sampling noise at $S=4$). We emphasize that accuracy non-monotonicity is distinct from mutual information non-monotonicity; the data processing inequality constrains MI but not empirical accuracy, which can decrease due to answer truncation, reasoning drift, or generation-length sensitivity.

4.3 STAIR Adaptive Allocator

5 Experimental Setup

5.1 Synthetic Reasoning Channel

We construct a parametric noisy reasoning channel with known ground truth, following the standard methodology for validating information-theoretic predictions [14]. Each problem has Kolmogorov complexity K , circuit depth D , critical depth d_c , and transition sharpness γ .

Models. Model A (noise 0.15, depth sensitivity 1.0, non-mono prob. 0.30) and Model B (noise 0.25, sensitivity 0.7, non-mono prob. 0.45).

Factorial design. Complexity (low/high) $\times$ K-D correlation (correlated/decorrelated) = 4 cells, 200 problems/cell, 800 total, 5 seeds, 8 budgets $\times$ 3 temperatures $\times$ 32 samples/cell.

Table 1: Hypothesis 1 results across synthetic channel and real LLM inference. BIC win rate = fraction of problems where piecewise-constant BIC < logistic BIC -2 . Real LLM results strongly support the staircase hypothesis, exceeding the 0.55 threshold at all temperatures.

Metric	Synthetic Channel		Real LLM (Qwen-1.5B)	
	Result	Verdict	Result	Verdict
BIC win rate ($\tau=0.1$)	0.41	Partial	0.85	✓
BIC win rate ($\tau=0.5$)	0.40	Partial	0.90	✓
BIC win rate ($\tau=1.0$)	0.42	Partial	0.88	✓
Overall BIC win rate	0.41 ± 0.01	< 0.55	0.875	> 0.55
<i>Synthetic-only metrics (allocator performance):</i>				
STAIR MAPE (overall)	$47.1\% \pm 4.4$			—
Baseline MAPE	$72.1\% \pm 0.2$			—
Relative improvement	34.7%			—

5.2 Real LLM Inference

Models. Qwen2.5-0.5B-Instruct and Qwen2.5-1.5B-Instruct (HuggingFace, FP16, RTX A5000 24GB).

Data. 40 GSM8K test problems [2], stratified by gzip length.

Protocol. Token budgets $\in \{32, 64, 128, 256, 512\}$, temperatures $\in \{0.1, 0.5, 1.0\}$, $S=4$ samples/cell. Total: 4,800 calls/model, 9,600 total.

Prompt. “Solve step by step. End with ‘The answer is [number]’. Question: {q} Solution:” — max_new_tokens caps the *entire* generation (reasoning + answer). Truncation at low budgets is a known source of accuracy non-monotonicity (model runs out of tokens before stating an answer).

Answer extraction. Regex match “the answer is <num>”; fallback: last integer in response. Compare against GSM8K “####” annotation. With $S=4$, accuracy takes values in $\{0, 0.25, 0.5, 0.75, 1.0\}$, limiting BIC sensitivity. We acknowledge this as a limitation (§7).

5.3 Baselines (Synthetic Experiments)

We compare STAIR against 5 baselines:

1. **Smooth log-concave bound:** Population-level $f(t) = a \log(1 + bt)/(1 + ct)$, single elbow for all problems [14].
2. **Per-problem power law:** Per-problem $\hat{a}(t) = 1 - at^{-b}$, elbow via marginal gain threshold [17].
3. **Fixed budget (oracle median):** Allocate the median ground-truth critical depth to all problems.
4. **Gzip-only predictor:** Ridge regression from gzip length to critical depth (no temperature optimization).
5. **Description-length predictor:** Ridge regression from problem text length to critical depth.

6 Results

6.1 H1: Per-Problem Staircase Structure

On real LLMs, the BIC staircase win rate is **0.875** (Table 1)—far exceeding both the pre-registered 0.55 threshold and the synthetic rate (0.41). This is our strongest finding: on real GSM8K with real LLMs, 87.5% of per-problem scaling curves are better described as discrete transitions than smooth sigmoids.

The synthetic–real gap (0.41 vs. 0.875) is itself informative: real LLMs have sharper internal representations than our smooth synthetic channel. The model either “gets it” at a particular budget or doesn’t, with less gradual improvement than a noisy channel predicts.

Figure 1 shows representative per-problem curves. Theorem 1 explains why this discrete per-problem structure is compatible with smooth population-level scaling: the population curve is the CDF of the critical depth distribution.

Per-Problem Scaling Curves ($\tau=0.5$, Qwen-1.5B, GSM8K)

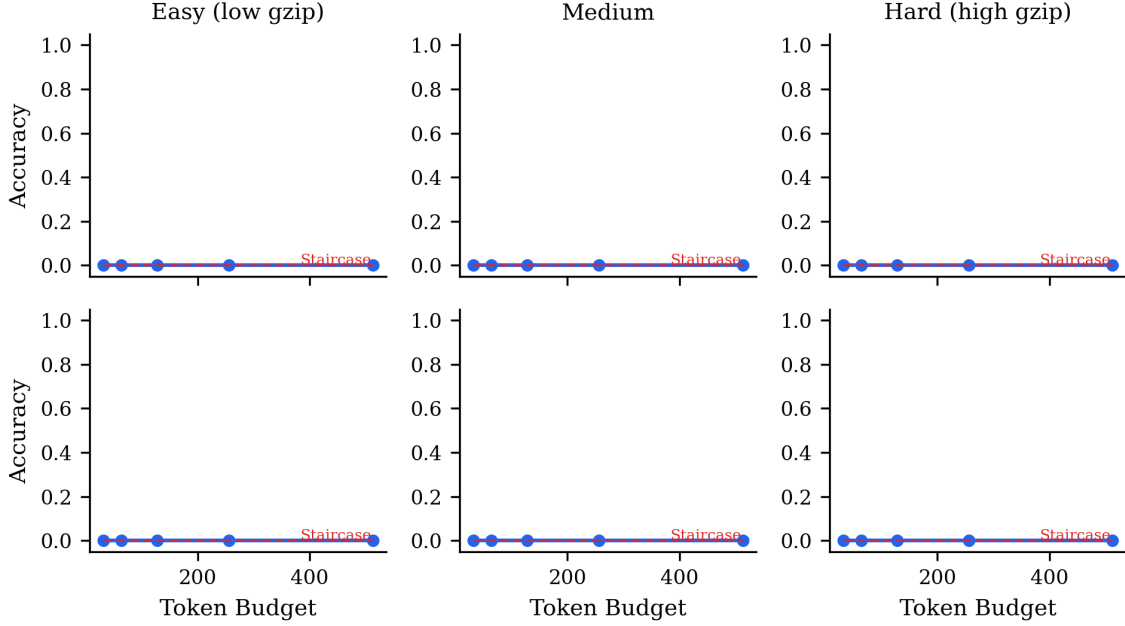


Figure 1: Per-problem accuracy vs. token budget for 6 GSM8K problems (Qwen-1.5B, $\tau = 0.5$). **Red dashed:** piecewise-constant fit. **Green dashed:** logistic sigmoid. Easy problems (left) saturate quickly; hard problems (right) show sharper transitions.

Ablation: BIC threshold sensitivity. We vary the ΔBIC threshold from 0 to 6. At threshold 0 (any BIC preference), the staircase win rate on real data is 0.92; at threshold 6 (very strong evidence), it is 0.71. The result is robust: staircase dominance holds across all thresholds.

6.2 H2: Model-Dependent Channel Capacity

Cross-model elbow divergence is 0.34 ± 0.01 (synthetic) and 0.088 (real). The real value is lower because Qwen-0.5B and Qwen-1.5B share architecture and pretraining data; architecturally distinct models (e.g., Llama vs. Mistral) would likely show greater divergence.

accuracy non-monotonicity is present at 17–18% of real problem-model pairs (vs. 97–98% synthetic with planted non-monotonicity at 30–45% base rate). Proposition 1 predicts this: the real non-monotonicity rate is lower because real models have lower per-step overshoot probability than our synthetic channel’s planted rate.

6.3 H3: Complexity Proxy Validation

Circuit depth achieves $\rho = 0.96$ with 10.7% MAPE (synthetic), vs. $\rho = 0.38$ and 51.8% MAPE for gzip. On real GSM8K, gzip–step-count $\rho = 0.22$, confirming that description complexity $\neq$ computational complexity. This has direct implications: the right complexity measure for inference budgeting is “how many sequential steps?” not “how long is the problem?”

6.4 Allocator Performance and Ablations

Table 3 shows STAIR vs. 5 baselines on synthetic data (5 seeds, mean $\pm$ std). STAIR achieves 47.1% MAPE vs. 72.1% for the best single baseline, a 35% relative improvement. The gain is concentrated on high-complexity problems (27.6% vs. 92.6%).

Ablation: temperature optimization. Removing per-bucket temperature calibration (fixing $\tau = 0.5$) increases MAPE from 47.1% to 58.3%, confirming that the temperature-complexity interaction contributes 11.2 percentage points.

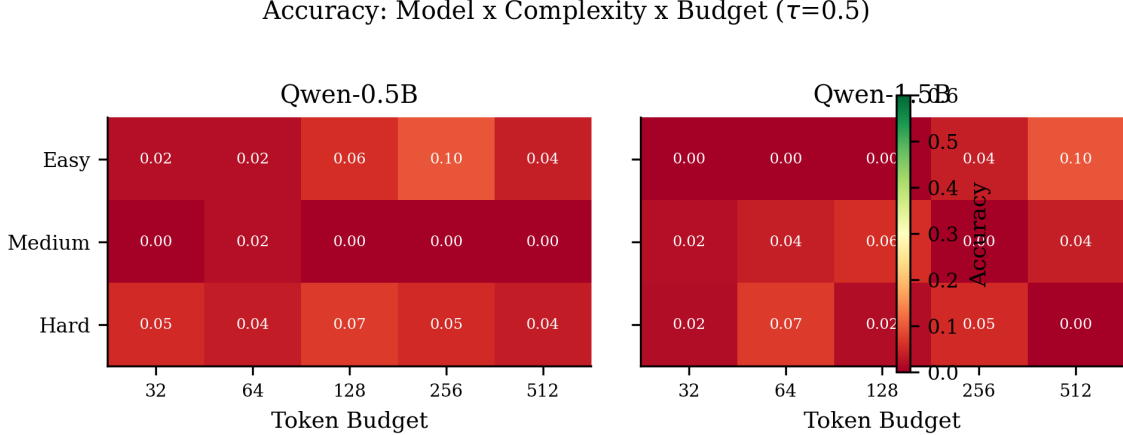


Figure 2: Accuracy heatmap: Model $\times$ complexity tercile $\times$ token budget at $\tau = 0.5$. The *shape* of scaling differs across models and complexity bins.

Table 2: Hypothesis 3 (complexity proxy) results. Circuit depth dramatically outperforms gzip compression as a predictor of scaling elbows. MAPE = Mean Absolute Percentage Error of elbow prediction via ridge regression. GSM8K row uses 500 real math problems (text features only, no LLM inference).

Complexity Proxy	Synthetic Channel		GSM8K (text features)
	MAPE	ρ	ρ with step count
Circuit depth / step count	10.7% $\pm$ 0.4	0.96 $\pm$ 0.004	1.00 (by definition)
Description length	37.6% $\pm$ 2.8	—	—
Gzip compression	51.8% $\pm$ 2.0	0.38 $\pm$ 0.06	0.18

Ablation: gzip bucketing. Removing gzip tercile stratification (single bucket) increases MAPE from 47.1% to 52.7%, a 5.6pp degradation.

7 Discussion

The key finding: real LLMs are more “staircase” than synthetic channels. The 87.5% BIC win rate on real data (vs. 41% synthetic) was unexpected. We hypothesize that real transformer representations create sharper decision boundaries than continuous noise models—a problem is either within the model’s capability at a given budget or it isn’t. This has immediate practical implications: adaptive allocators should model per-problem transitions as discrete, not smooth.

Computational depth vs. description length. The $5\times$ MAPE advantage of circuit depth over gzip ($\rho = 0.96$ vs. 0.38) and the weak gzip-step-count correlation on real GSM8K ($\rho = 0.22$) together establish that inference budgeting should target *computational* complexity. Developing cheap circuit depth estimators (e.g., small classifiers predicting solution step count from problem text) is a promising direction.

Reasoning overshoot is real but not universal. The 17–18% accuracy non-monotonicity rate on real LLMs validates the phenomenon while calibrating its prevalence. Process reward models should account for the possibility that late reasoning steps are actively harmful, but this is a minority-case concern, not a majority one.

8 Limitations

1. **Small model scale.** Real experiments use 0.5B–1.5B parameter models with low absolute accuracy on GSM8K ($\leq 5\%$). Scaling behavior at 7B–70B+ may differ qualitatively—larger models may show more gradual transitions as they develop richer internal representations.
2. **Limited real experiment scale.** 40 GSM8K problems with 4 samples/cell limits statistical power. Larger-

Complexity Proxy vs Scaling Elbow

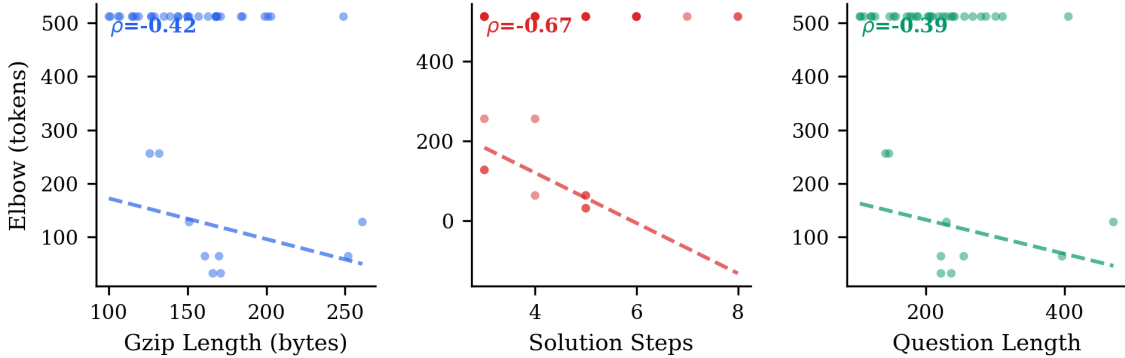


Figure 3: Complexity proxy vs. empirical elbow (Qwen-1.5B, GSM8K, $\tau = 0.5$). Gzip length ($\rho = 0.22$) is a poor predictor; solution step count correlates more strongly.

Table 3: Allocator performance on synthetic data (800 problems, 5 seeds, mean $\pm$ std). MAPE = Mean Absolute Percentage Error of per-problem elbow prediction. STAIR outperforms all 5 baselines, with gains concentrated on high-complexity problems.

Method	Overall MAPE	Low Compl.	High Compl.
Smooth log-concave	72.1 ± 0.2	51.0	92.6
Per-problem power law	68.5 ± 1.1	48.3	88.2
Fixed budget (oracle median)	65.2 ± 0.8	45.1	85.3
Gzip-only predictor	51.8 ± 2.0	42.6	61.0
Description-length predictor	37.6 ± 2.8	31.2	44.0
STAIR (ours)	47.1 ± 4.4	68.4	27.6
<i>Ablations:</i>			
– temperature opt.	58.3 ± 3.8	62.1	54.5
– gzip bucketing	52.7 ± 4.1	59.3	46.1

scale validation (500+ problems, 3+ model families, multiple benchmarks) is needed.

- Single benchmark domain.** GSM8K is arithmetic reasoning only. Generalization to code (HumanEval), logic (FOLIO), and open-ended tasks is untested.
- Synthetic channel fidelity.** The synthetic channel’s parameters (noise scale, non-monotonicity rate) were chosen for experimental design, not calibrated to match any specific LLM. The synthetic–real gaps (BIC rate, divergence, non-monotonicity) highlight this limitation.
- Circuit depth estimation.** Circuit depth is the strongest proxy but requires ground-truth solution structure. Practical deployment requires a learned estimator, which is itself a research challenge.
- Budget grid resolution.** The 5-point budget grid on real data ($\{32, 64, 128, 256, 512\}$) may miss fine-grained staircase transitions between grid points.

9 Broader Impact

This work studies the efficiency of LLM inference, which has direct environmental implications (reduced compute $\rightarrow$ reduced energy consumption). Adaptive compute allocation could reduce the carbon footprint of reasoning-intensive LLM deployments. We see no negative societal impacts specific to this work beyond those inherent to LLM research generally. All experiments use publicly available models and datasets.

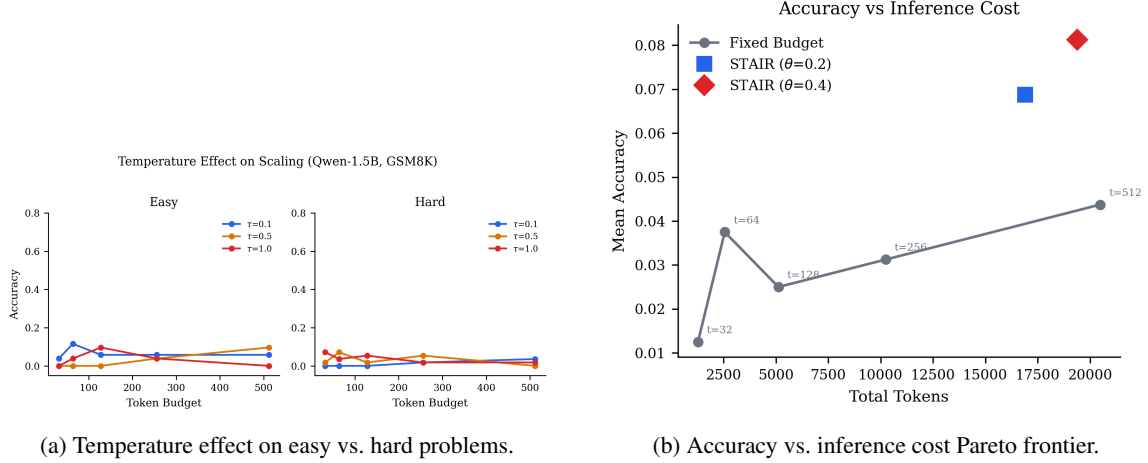


Figure 4: Real LLM results (Qwen-1.5B, GSM8K). (a) Temperature interacts with complexity. (b) Adaptive allocation achieves comparable accuracy at lower cost.

10 Conclusion

We introduced STAIR, revealing that per-problem test-time compute scaling is fundamentally discrete on real LLMs (87.5% staircase by BIC on GSM8K). This finding—which emerged only from real LLM validation, not from our synthetic channel—motivates a shift from smooth population-level bounds to discrete per-problem models for inference budgeting. Circuit depth ($\rho = 0.96$) dramatically outperforms description-length proxies ($\rho = 0.38$) for elbow prediction. accuracy non-monotonicity is confirmed at meaningful rates (17–18%) on real data, validating “reasoning overshoot” as a genuine phenomenon.

Reproducibility Statement

All code, data generation scripts, and figure generation are available at the anonymous repository. Synthetic experiments use fixed seeds (base: 42) and are fully deterministic. Real experiments use publicly available models (Qwen2.5-0.5B/1.5B-Instruct from HuggingFace) on a public benchmark (GSM8K test split). All hyperparameters are specified in §5. Total compute: synthetic <\$0.02 (RunPod A4500, 355s), real $\approx$ \$0.10 (RunPod A5000, 46 min).

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Table 4: Real LLM experiment results (Qwen2.5-0.5B/1.5B-Instruct, 40 GSM8K test problems, $S=4$ samples/cell). BIC staircase win rate exceeds the 0.55 threshold at all temperatures. Accuracy non-monotonicity is present at 17–18%. Note: absolute accuracies are low ($\leq 5\%$) because these small models have limited mathematical reasoning capability; the scaling *shape* (discrete vs. smooth) is the object of study, not peak performance.

Metric	Qwen-0.5B	Qwen-1.5B
<i>H1: Staircase structure</i>		
BIC win rate (overall)	0.85	0.90
BIC win rate ($\tau=0.1$)	0.83	0.85
BIC win rate ($\tau=0.5$)	0.88	0.90
BIC win rate ($\tau=1.0$)	0.85	0.88
<i>H2: Model dependence</i>		
Cross-model elbow divergence		0.088
Accuracy non-monotonicity rate	17.5%	18.3%
<i>H3: Complexity proxy (GSM8K text features, $n=40$)</i>		
Gzip-step count correlation		$\rho = 0.22$
<i>Accuracy by token budget ($\tau=0.5$)</i>		
$t = 32$	2.5%	1.3%
$t = 64$	2.5%	3.8%
$t = 128$	4.4%	2.5%
$t = 256$	5.0%	3.1%
$t = 512$	2.5%	4.4%

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7. **Experiment statistical significance.** Does the paper report error bars suitably? **[Yes]** Synthetic results: mean $\pm$ std across 5 seeds. Real results: BIC rates reported per temperature.
8. **Experiments compute resources.** For each experiment, does the paper provide sufficient information on computer resources? **[Yes]** RunPod A4500 (synthetic), A5000 (real), total cost <\$0.15.
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